

Linear Algebra and Calculus Lecture 7

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Linear Independence

Definition. A set of vectors $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is called **linearly** _____ if the vector equation

$$x_1v_1 + x_2v_2 + \dots + x_pv_p = 0$$

only has solution _____. It is called **linearly** _____ if the previous vector equation has nontrivial solutions.

Proposition. If A is an $m \times n$ matrix with columns $a_1, \dots, a_n \in \mathbb{R}^m$, then the vectors $a_1, \dots, a_n$ are linearly independent if and only if the matrix equation $Ax = 0$ has no _____.

Question 1. Let $v \in \mathbb{R}^n$. Is v necessarily linearly independent?

Exercise. Determine if the following sets of vectors are linearly independent.

- $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$.
- $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$.

Proposition. Two non-zero vectors v_1, v_2 are linearly independent if and only if v_1 is not a scalar multiple of v_2 .

Theorem. (Characterization of Linearly Independent Sets) An indexed set $S = \{v_1, \dots, v_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $v_1 \neq 0$, then some v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \dots, v_{j-1}$.

Theorem. If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is linearly dependent if _____.

Example. Any matrix with more _____ than _____ will have linearly dependent columns.

Theorem. If a set $S = \{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ contains _____, then the set is linearly dependent.

Linear Transformations

Definition. A **transformation** (or **map** or **function**) $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a rule that assigns to each vector $x \in \mathbb{R}^n$ a vector $T(x) \in \mathbb{R}^m$.

The definitions of domain, codomain, range, image, etc are the same as in calculus. In this section we will focus on *matrix maps*, that is, linear maps T in which $T(x) = \underline{\quad}$ for some matrix A of size $m \times n$.

Example. Let

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}.$$

Let's determine if c is in the range of the map T given by $T(x) = Ax$.

The vector c is in the range of T if c is the image of some $x \in \mathbb{R}^2$, that is, if $c = T(x)$ for some x . This is just another way of asking if the system $Ax = c$ is consistent. To find the answer, row reduce the augmented matrix:

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

The third equation, $0 = -35$, shows that the system is inconsistent. So c is not in the range of T .

Definition. A transformation (or mapping) T is **linear** if

1. $T(u + v) = \underline{\hspace{2cm}}$, for all u, v in the domain of T ;
2. $T(cu) = \underline{\hspace{2cm}}$ for all scalars c and all u in the domain of T .

Remark. Every matrix transformation is a linear transformation (since $A(u + v) = Au + Av$ and $A(cu) = cAu$), but not every linear transformation is a matrix transformation.

Remark. If T is a linear transformation, then $T(0) = 0$, and $T(cu + dv) = cT(u) + dT(v)$, for all vectors u, v in the domain of T and all scalars c, d .

Of course, this can be generalized to

$$T(c_1v_1 + \cdots c_pv_p) = \underline{\hspace{2cm}}.$$

In fact, this last conditions summarizes the two conditions of the definition, so it is enough to prove linearity.

Example. Given a scalar r , define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = rx$. T is called a **contraction** when $0 \leq r \leq 1$ and a **dilation** $r > 1$. Let's prove it is a linear transformation. Let $u, v \in \mathbb{R}^2$ and let c, d be scalars. Then

$$T(cu + dv) = \underline{\hspace{2cm}} = cT(u) + dT(v).$$

Exercise. Let T be a matrix map given by $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find the images under T of $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $u + v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.